LONGITUDINAL BASELINE MODELING AND ADAPTIVE PERSONALIZATION

Personalized Baseline Establishment

The wearable longitudinal penile physiology monitoring device may establish individualized physiologic baselines derived from repeated measurements collected across extended monitoring periods. Baseline modeling may characterize typical rigidity behavior, physiologic variability, temporal response patterns, and recovery dynamics unique to a given user rather than relying on population averages or fixed threshold definitions.

Baseline establishment may occur automatically through passive data collection, guided calibration sessions, or hybrid approaches combining user interaction and algorithmic learning. Baselines may continuously evolve as additional longitudinal data becomes available.

Adaptive Personalization Framework

Adaptive personalization algorithms may modify interpretation parameters based on learned individual characteristics including typical erection onset speed, rigidity stability patterns, nocturnal activation frequency, thermal response behavior, and motion context trends. Personalization may improve measurement accuracy by accounting for anatomical variability, behavioral differences, and environmental influences.

Adaptive systems may dynamically adjust classification thresholds, weighting factors, or confidence metrics without requiring manual recalibration. Machine learning models, statistical adaptation methods, or rule-based optimization techniques may be employed to refine individualized interpretation over time.

Deviation Detection and Trend Monitoring

Longitudinal baselines may enable detection of deviations from typical physiologic behavior. Trend monitoring may identify gradual changes, short-term fluctuations, recovery patterns, adaptation responses, or anomalous events relative to a user’s historical profile. Deviation detection may operate continuously or during defined analysis intervals.

Trend analysis may incorporate rolling averages, temporal segmentation, pattern comparison, or probabilistic anomaly detection techniques capable of identifying meaningful physiologic changes while minimizing false interpretation.

Cross-Temporal and Comparative Analysis

Baseline models may support comparison between different time periods, environmental conditions, operational modes, or user-defined contexts. Comparative analysis may evaluate short-term versus long-term behavior, recovery trajectories following physiologic events, and adaptation to lifestyle or environmental changes.

Privacy-Preserving Personal Models

Personalized baseline models may be maintained locally on device, within user-controlled computing environments, or within secure distributed processing systems. Personal models may remain private to the user while still enabling aggregated learning frameworks or anonymized system-level improvements when permitted by configuration.

The adaptive personalization embodiments described herein are illustrative and non-limiting and encompass present and future computational methods capable of learning individualized physiologic patterns through longitudinal wearable sensing.